<u>Lem 1</u> Let S be a closed subset of a complete metric space (M, d). Then S is complete wrt the restriction of d (i.e., $(S, d|_{S\times S})$ is complete).

<u>Pf</u> Any Cauchy sequence (x_n) in S is also a Cauchy sequence in M, so converges to a limit $x \in M$. But S is a closed subset of M, so $x \in S$. $\square$

<u>Lem 2</u> (integral formulation of IVP)

Let $E^{\text{open}} \subset \mathbb{R}^n$ and $f: E \to \mathbb{R}^n$ be continuous. Then $u: I \to \mathbb{R}^n$ solves IVP "$\dot{x} = f(x),\ x(0) = x_0$"

$\iff$ 1. u is continuous,

2. $\forall t \in I: \ u(t) = x_0 + \int_0^t f(u(s))\,ds.$

pf $\Rightarrow$) Assume $u: I \to \mathbb{R}^n$ solves IVP. Then u is diff., hence cont. Since f and u are continuous, so is $t \mapsto \dot{u}(t) = f(u(t))$. Thus, FTC (pt II)

$$\Rightarrow \quad u(t) = u(0) + \int_0^t \dot{u}(s)\,ds, \quad \text{or}$$

$$u(t) = x_0 + \int_0^t f(u(s))\,ds. \quad \checkmark$$

$\Leftarrow$) Assume $u: I \to \mathbb{R}^n$ is cont. and that

$$u(t) = x_0 + \int_0^t f(u(s))\,ds \quad \forall t \in I.$$

Then $u(0) = x_0$. Since f, u, and hence also $s \mapsto f(u(s))$ is continuous, the FTC (pt I)

$$\Rightarrow \quad \dot{u}(t) = f(u(t)). \quad \checkmark$$

$\square$

Picard-Lindelöf Thm Let $E^{\text{open}} \subset \mathbb{R}^n$ and $f: E \to \mathbb{R}^n$ be locally Lipschitz. Fix any $x_0 \in E$. Then $\exists\, \delta, \varepsilon > 0$ s.t. for any $y_0 \in \overline{B}_\delta(x_0)$, there exists a unique solution on $[-\varepsilon, \varepsilon]$ to the IVP "$\dot{x} = f(x)$, $x(0) = y_0$".

Rmk $\varepsilon > 0$ is independent of $y_0 \in \overline{B}_\delta(x_0)$.

Pf Fix $\delta > 0$ s.t. $B := \overline{B}_{2\delta}(x_0) \subset E$.

Since $\underbrace{B \text{ is cpct}}_{\text{by Heine-Borel}}$, $\|f\|$ is bounded by some $M > 0$ on B, and f is L-Lipschitz on B (by lemma from last time) for some $L > 0$.

Choose $\varepsilon > 0$ s.t. $\varepsilon < \min\left\{\frac{\delta}{M}, \frac{1}{L}\right\}$.

Define $S := C^0([-\varepsilon, \varepsilon], B) \subset C^0([-\varepsilon, \varepsilon], \mathbb{R}^n)$ equipped w/ supremum metric. Note that S is complete metric space (since it is a closed subspace of a Banach space; Lemma 1).

Given any $y_0 \in \overline{B}_\delta(x_0)$, define Picard operator $T_{y_0} : S \to S$ by

$$T_{y_0}(u)(t) := y_0 + \int_0^t f(u(s))ds.$$

This is well-defined (i.e., $T_{y_0}(S) \subset S$). To see why, fix any $u \in S$. Then by defn of S, $u(s) \in B$ and hence $\|f(u(s))\| \leq M$ for all $s \in [-\varepsilon, \varepsilon]$. Thus, $\forall t \in [-\varepsilon, \varepsilon]$, $u \in S$,

$$\begin{aligned} \|T_{y_0}(u)(t) - y_0\| &\leq \int_0^t \|f(u(s))\| ds \leq M|t| \\ &\leq M\varepsilon < M\frac{\delta}{M} = \delta, \end{aligned}$$

So $T_{y_0}(u)(t) \in B_\delta(y_0)$. The triangle inequality then yields $T_{y_0}(u)(t) \in B_{2\delta}(x_0) \subset B$ (since $y_0 \in \overline{B}_\delta(x_0)$). Thus, $T_{y_0}(u) \in S$ $\forall u \in S$.

We now prove that $T_{y_0} : S \to S$ is a contraction map.

$\forall u, v \in S$, $t \in [-\varepsilon, \varepsilon]$, have

$$\| T_{y_0}(u)(t) - T_{y_0}(v)(t) \| \le \operatorname{sign}(t) \int_0^t \| f(u(s)) - f(v(s)) \| \, ds$$

$$\le L \cdot \operatorname{sign}(t) \int_0^t \| u(s) - v(s) \| ds$$

$$\le L \varepsilon \| u - v \|_\infty$$

$\Rightarrow \| T_{y_0}(u) - T_{y_0}(v) \|_\infty \le L\varepsilon \| u - v \|_\infty$.

Since $\varepsilon < \frac{1}{L}$, $L\varepsilon < 1$, so <u>T_{y_0} is a contraction map on S</u>

(wrt sup metric). Thus, T_{y_0} has a <u>unique</u> fixed pt

$u_{y_0} \in S$. So by Lemma 2, u_{y_0} is the unique

soln on $[-\varepsilon, \varepsilon]$ to "$\dot{x} = f(x)$, $x(0) = y_0$".

Since $y_0 \in \overline{B}_\delta(x_0)$ was arbitrary, have shown that $\forall y_0 \in \overline{B}_\delta(x_0)$ $\exists!$ soln to "$\dot{x} = f(x)$, $x(0) = y_0$" on $[-\varepsilon, \varepsilon]$ (w/ ε independent of y_0). $\square$

Corollary Previous (P-L) thm remains true w/ $t=0$ replaced by $t=t_0$, $[-\varepsilon, \varepsilon]$ replaced by $[t_0-\varepsilon, t_0+\varepsilon]$.

pf Let $\delta, \varepsilon > 0$ satisfy the conclusion of P-L thm.

Then $\forall y_0 \in \overline{B}_\delta(x_0)$, $\exists!$ soln $x_{y_0}(t)$ to "$\dot{x} = f(x)$, $x(0) = y_0$" on $[-\varepsilon, \varepsilon]$.

Hence $t \longmapsto x_{y_0}(t - t_0)$ is soln on $[t_0 - \varepsilon, t_0 + \varepsilon]$ to "$\dot{y} = f(y)$, $y(t_0) = y_0$".

Now suppose $t \mapsto y_1(t), y_2(t)$ are solns to the latter IVP on $[t_0-\varepsilon, t_0+\varepsilon]$. Then $x_1(t) := y_1(t+t_0)$ and $x_2(t) := y_2(t+t_0)$, then the x_i both satisfy the IVP "$\dot{x} = f(x), x(0) = y_0$", so the uniqueness portion of P–L $\Rightarrow$ $x_1 = x_2$ and hence $y_1 = y_2$. $\square$